

NANO 2314 Midterm — after-exam review

The midterm was given on Monday, October 5, 2026. Everyone got the same booklet — there were no scrambled versions — so the answer letters below are the same for the whole class.

Questions 1–5 are multiple choice (1 point each, filled bubble), and so is part (a) of Question 6. Questions 6–8 are free response, graded with partial credit, for a total of **20 points**. Questions 9 and 10 are **extra credit**, 1.5 points each; they only add to your score.

This is the printable version: every solution is shown under its question, and the correct multiple-choice answer is highlighted in green. The interactive page, with the solutions hidden so you can try each question first, is at <https://tinyurl.com/N2314-F26-Midterm-review>.

The exam pulls together the first half of the course: units inside exponentials and cosines (Questions 1–5), the shifted variable $T - T_a$ and its solution (Question 6), the series argument for a derivative (Question 7), and reading and critiquing a real result (Question 8).

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1. (1 pt) Questions 1–5 refer to the damped oscillation

$$x(t) = A e^{-\gamma t} \cos(\omega t + \delta),$$

where time t is measured in seconds and the displacement x is measured in mm.

The appropriate unit for the **decay rate** γ is

- A. seconds
- B. inverse seconds —correct
- C. mm
- D. inverse mm
- E. unitless

Solution

Correct answer: B. inverse seconds

Why: The exponent γt must be a pure number (Question 2), and t is in seconds, so

$$[\gamma] = \frac{1}{[t]} = \text{s}^{-1}.$$

It is the same as r in e^{-rt} on Quiz 4.

Watch out: Choice A is the unit of t — and of the decay *time* $\tau = 1/\gamma$ — not of the rate; it would make γt come out in s^2 , which cannot sit in an exponent. Choice C: the length lives in the amplitude A , outside the exponential. Choice D has the right *form* for a decay constant (one over the variable in the exponent) but the wrong variable: inverse mm would go with decay in space, $e^{-x/\lambda}$. Choice E: only the *product* γt is unitless; γ itself has to cancel the second.

2. (1 pt) The appropriate unit for the **product** γt in $e^{-\gamma t}$ is

- A. seconds
- B. inverse seconds
- C. mm
- D. inverse mm
- E. **unitless —correct**

Solution

Correct answer: E. unitless

Why: An exponent must be a pure number. From the series

$$e^{-\gamma t} = 1 - \gamma t + \frac{1}{2}(\gamma t)^2 - \dots$$

you cannot add the pure number 1 to something measured in seconds. In the product the seconds cancel: $(1/s)(s) = 1$. Same idea as rt on Quiz 4 and ωt on Quiz 5.

Watch out: Choice A is the unit of t alone and Choice B the unit of γ alone (Question 1); multiplying them is exactly what removes the second. Choices C and D are lengths, and nothing in the exponent involves a length — the mm lives in A .

3. (1 pt) Which of the following is a widely used “unit” that is actually unitless?

- A. seconds
- B. inverse seconds
- C. mm
- D. inverse mm
- E. **radians —correct**

Solution

Correct answer: E. radians

Why: A radian is arc length divided by radius (mm/mm), a pure number. That is why the cosine’s argument $\omega t + \delta$ can be “in radians” and still be dimensionless, and why rad/s is the same as s^{-1} .

Watch out: Choices A–D are all real dimensions: time, inverse time, length and inverse length.

4. (1 pt) The parameter ω is called _____ and its unit should be _____.

- A. period; inverse seconds
- B. phase constant; seconds
- C. frequency; unitless
- D. **angular frequency; radians per second —correct**
- E. phase constant; radians

Solution

Correct answer: D. angular frequency; radians per second

Why: ωt must be an angle in radians and t is in seconds, so

$$[\omega] = \frac{\text{rad}}{\text{s}} (= \text{s}^{-1}).$$

ω is the **angular frequency**, $\omega = 2\pi f = 2\pi/T$. Same as Quiz 5, Question 2.

Watch out: Choice A: ω is not the period ($\omega = 2\pi/T$), and a period is measured in seconds, not inverse seconds. Choices B and E describe δ (Question 5). Choice C: the ordinary frequency is $f = 1/T$; ω is the *angular* frequency, and neither is unitless — a unitless ω would leave seconds in ωt .

5. (1 pt) The parameter δ is called _____ and its unit should be _____.

- A. period; inverse seconds
- B. phase constant; seconds
- C. frequency; unitless
- D. angular frequency; radians per second
- E. **phase constant; radians —correct**

Solution

Correct answer: E. phase constant; radians

Why: δ is **added** to the angle ωt inside the cosine, so it is itself an angle, in radians. It is the **phase constant**: it sets where in the cycle the oscillation is at $t = 0$, since $x(0) = A \cos \delta$.

Watch out: Choice B has the right name but the wrong unit: only like quantities can be added, and radians plus seconds has no meaning. Choice A: δ is neither a time nor an inverse time. Choice C: δ does not set how fast the motion repeats. Choice D describes ω (Question 4).

6. (5 pts) A **cold** drink sits in a room with an ambient temperature of $T_a = 76^\circ\text{F}$. Initially (at $t = 0$), the drink's temperature T is 36°F . The drink's temperature as a function of time, $T(t)$, follows the first-order differential equation

$$\frac{d}{dt}[T - T_a] = -r[T - T_a],$$

where $r > 0$ is the decay rate.

- a. **(1 pt)** In the situation described here, the sign of $T - T_a$ is (fill in one bubble) A. positive, B. negative, C. undefined.

Solution — part (a)

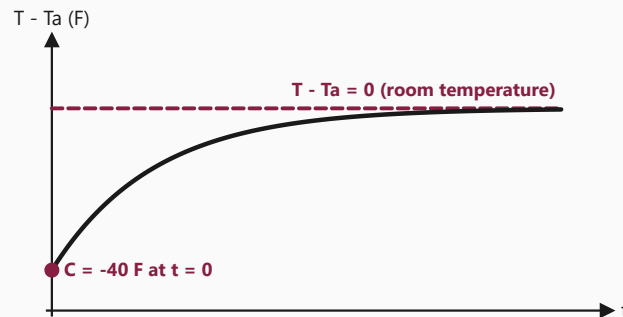
Negative — B. At $t = 0$, $T - T_a = 36 - 76 = -40^\circ\text{F}$. The drink warms toward 76°F but never passes it, so $T - T_a$ stays negative the whole time.

- b. **(2 pts)** Sketch $T - T_a$ vs. t (not T vs. t). Label C = the initial temperature deviation $T - T_a$ at $t = 0$.

Solution — part (b)

Put $T - T_a$ on the vertical axis and t on the horizontal axis. The curve starts **below** the t -axis at $C = T(0) - T_a = -40^\circ\text{F}$

and rises toward 0 — steepest at $t = 0$, then flattening out — approaching the t -axis without ever crossing it. Label C at that starting point. This is the mirror image of Quiz 4's hot object, where C was positive and the curve fell toward 0 from above. Two common slips: plotting T itself (rising from 36 to 76), and calling C the initial *temperature* $T_0 = 36^\circ\text{F}$ — C is the initial *deviation* from room temperature.



$T - T_a = -40 e^{-rt}$ (in $^\circ\text{F}$), drawn against t . The dashed line is $T - T_a = 0$, the drink at room temperature. The curve starts at $C = -40^\circ\text{F}$, is steepest at $t = 0$, and rises toward the dashed line without crossing it.

- c. **(2 pts)** Write a legitimate solution of the differential equation above, and **verify** it by substitution and differentiation.

Solution — part (c)

A solution:

$$T - T_a = C e^{-rt}, \quad \text{i.e.} \quad T(t) = T_a + C e^{-rt},$$

with any constant C (for this drink $C = -40^\circ\text{F}$, so $T = 76 - 40 e^{-rt}$).

Verify. Left side:

$$\frac{d}{dt}[C e^{-rt}] = -r C e^{-rt}.$$

Right side:

$$-r[T - T_a] = -r C e^{-rt}.$$

The two sides are equal, so the solution is legitimate. A verification has to *differentiate*: writing the equation again, or plugging in numbers at $t = 0$, does not check anything. And $T = C e^{-rt}$ on its own is **not** a solution for T — the T_a is what makes the drink approach room temperature.

7. **(6 pts)** Just as e^{rt} can be written as $e^{rt} = 1 + (rt) + (rt)^2/2! + (rt)^3/3! + (rt)^4/4! + \dots$, $\cos(\omega t)$ can be written as

$$\cos(\omega t) = 1 - \frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} - \frac{(\omega t)^6}{6!} + \dots$$

- a. **(2 pts)** Take the **first** derivative with respect to time t of each of the four terms

$$1, \quad -\frac{(\omega t)^2}{2!}, \quad \frac{(\omega t)^4}{4!}, \quad -\frac{(\omega t)^6}{6!}.$$

Solution — part (a)

Expand each power, $(\omega t)^n = \omega^n t^n$, and use the power rule with every letter except t held constant:

$$\frac{d}{dt}1 = 0, \quad \frac{d}{dt}\left[-\frac{\omega^2 t^2}{2!}\right] = -\frac{2 \omega^2 t}{2!} = -\omega^2 t,$$

$$\frac{d}{dt}\left[\frac{\omega^4 t^4}{4!}\right] = \frac{4 \omega^4 t^3}{4!} = \frac{\omega^4 t^3}{3!}, \quad \frac{d}{dt}\left[-\frac{\omega^6 t^6}{6!}\right] = -\frac{6 \omega^6 t^5}{6!} = -\frac{\omega^6 t^5}{5!}.$$

Watch the ω 's: $(\omega t)^2 = \omega^2 t^2$, so its derivative is $2\omega^2 t$, not $2\omega t$. And bring the exponent down: the derivative of $\omega^4 t^4/4!$ is not $\omega^4 t^3/4!$. Unsimplified factorials ($4\omega^4 t^3/24$) are perfectly fine.

b. (2 pts) Take the **second** derivative with respect to t of each of the same four terms.

Solution — part (b)

Differentiate the results of (a) once more:

$$0, \quad -\omega^2, \quad \frac{3\omega^4 t^2}{3!} = \frac{\omega^4 t^2}{2!}, \quad -\frac{5\omega^6 t^4}{5!} = -\frac{\omega^6 t^4}{4!}.$$

c. (2 pts) Using your answer from (b), make a case that $\frac{d^2}{dt^2}\cos(\omega t) = -\omega^2\cos(\omega t)$.

Solution — part (c)

Add the results of (b) and **factor out** $-\omega^2$:

$$-\omega^2 + \frac{\omega^4 t^2}{2!} - \frac{\omega^6 t^4}{4!} + \dots = -\omega^2 \left[1 - \frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} - \dots \right].$$

The bracket **is the series for** $\cos(\omega t)$, so

$$\frac{d^2}{dt^2}\cos(\omega t) = -\omega^2\cos(\omega t).$$

Four terms in give three non-zero terms out, because the constant 1 differentiates to 0; the next term of the series, $+(\omega t)^8/8!$, supplies the next bracket term, so the pattern continues forever. The chain rule ($\cos \rightarrow -\omega\sin \rightarrow -\omega^2\cos$) gives the same answer, but the question asks you to *build* the case from your (b) — that is the point of the series argument (Quizzes 3 and 4).

8. (4 pts) You're supervising an inexperienced intern named Urotas Irome, who is quantifying the stiffness k of a cantilever. Urotas has measured the oscillation period T for 6 different samples with different masses. The dynamics of the cantilever is approximated by the undamped harmonic oscillator (a second-order differential equation without a damping term)

$$\frac{d^2}{dt^2} z(t) = -\omega^2 z(t),$$

where $\omega^2 = k/m$, and m is the mass of the sample cantilever.

a. **(2 pts)** As usual, $\omega = 2\pi/T$. What is T in terms of k and m ?

Solution — part (a)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{k/m}} = 2\pi\sqrt{\frac{m}{k}}.$$

One over the square root of a fraction is the square root of the **flipped** fraction. Check it: a heavier sample swings more slowly (longer T , as in Urotas's graph), and a stiffer cantilever faster. It is the twin of the pendulum's $T = 2\pi\sqrt{L/g}$ from Quiz 5.

b. **(2 pts)** Urotas has given you a graph of T (vertical axis) against m (horizontal axis): six data points at $m = 1, 2, \dots, 6$ with error bars, a square-root fit curve through them, tick numbers on both axes but no units, and a legend reading "Fit result: $k = 6.762026767$ ". What should you tell or advise Urotas?

Solution — part (b)

Four things are wrong with the *report*:

1. The vertical axis T has **no unit** (seconds? microseconds?).
2. The horizontal axis m has **no unit** (kg? ng?).
3. The fit result k has **no unit**.
4. $k = 6.762026767$ has **ten significant figures and no uncertainty**, although the data scatter and carry error bars of several percent. Report the fit uncertainty and round to it — something like $k = 6.8 \pm 0.3$, in its unit.

What is fine: the data do follow the $\sqrt{m}$ trend, and the points carry error bars. "The fit looks good" is not advice: the problems are the missing units and the unjustified precision. For reference, one consistent set of units is T in μs , m in ng and k in N/m. The unit of k is Question 9.

9. (1.5 pts) Extra credit. What could be an appropriate unit for the stiffness k in Question 8?

Solution

From $\omega^2 = k/m$:

$$k = m \omega^2, \quad [k] = \text{kg} \cdot \text{s}^{-2} = \text{kg/s}^2.$$

Since $\text{N} = \text{kg m/s}^2$, this is the same as N/m — force per unit deflection, as in Hooke's law $F = -kz$. Any mass unit over seconds squared (such as g/s^2) is an appropriate unit too. AFM data sheets quote cantilever stiffness in N/m (equivalently nN/nm).

10. (1.5 pts) Extra credit. Consider the differential equation

$$\frac{d^2}{dt^2} x(t) = -\omega^2 x(t).$$

Show that $x(t) = A \cos(\omega t) + x_0$, with $x_0 \neq 0$, is **NOT** a legitimate solution (e.g., by substitution and differentiation). How can you modify the differential equation to make it a legitimate solution?

Solution

Substitute. Left side — the constant x_0 differentiates to 0:

$$\frac{d^2}{dt^2} [A \cos(\omega t) + x_0] = -A \omega^2 \cos(\omega t).$$

Right side — the x_0 stays:

$$-\omega^2 [A \cos(\omega t) + x_0] = -A \omega^2 \cos(\omega t) - \omega^2 x_0.$$

The two sides differ by $\omega^2 x_0 \neq 0$, so it is **not** a solution. (x_0 drops out of the left side only — not out of both.)

Fix the equation, not the solution: measure the displacement from x_0 — the same move as $T - T_a$ in Question 6:

$$\frac{d^2}{dt^2} [x - x_0] = -\omega^2 [x - x_0], \quad \text{i.e.} \quad \frac{d^2 x}{dt^2} = -\omega^2 (x - x_0) = -\omega^2 x + \omega^2 x_0.$$

Now both sides equal $-A \omega^2 \cos(\omega t)$. Physically, the oscillation is about an equilibrium at x_0 instead of 0 — a mass hanging on a vertical spring, for example. Note the coefficient: the added term is $\omega^2 x_0$, not just x_0 .

Where to review each idea

Class results

14 exams were graded.

- **Midterm score (out of 20):** average 16.7, median 17.6
- **Multiple choice, Questions 1–5 and 6(a) (out of 6):** average 5.9 — nearly everyone got every one
- **Free response, Questions 6(b)–8 (out of 14):** average 10.8
- **Extra credit, Questions 9–10 (out of 3):** average 1.3

Where the class had the most trouble

These cost the most points. If you lost points here, start your review with them.

1. **Question 8(b) — the missing units.** Everyone caught that $k = 6.762026767$ has far too many digits, and most said k has no unit, but only about a third noticed that **neither axis has a unit**. A graph whose axes have no units cannot be read: T in seconds and T in microseconds are very different experiments. This is the same lesson as the intern on Quiz 5.
2. **Question 6(b) — plotting $T - T_a$, not T .** Every bubble in 6(a) was right ($T - T_a$ is negative), yet several sketches showed T rising from 36 to 76, or a curve above the axis. If $T - T_a$ is negative, its graph lives below the t -axis, and C is the starting value there, -40°F — not the starting temperature.
3. **Question 10 — fixing the equation.** Most who tried showed that x_0 breaks the equation, but only one wrote a correct modified equation. Adding x_0 alone is not enough — the term is $\omega^2 x_0$ — and changing the *solution* (dropping x_0 , adding a phase constant) does not answer the question.
4. **Question 7 — the details of the derivatives.** Hold ω constant and keep it: $(\omega t)^2$ is $\omega^2 t^2$. Bring the exponent down, keep the minus signs, and remember that the derivative of the constant 1 is 0. In (c), factor $-\omega^2$ out of *your* (b) — the chain rule alone does not use the series.
5. **Question 6(c) — a solution is a function of t .** Plugging numbers in at $t = 0$ gives one value, not a solution, and a verification has to differentiate.

Earlier quizzes to revisit, question by question

- **Q1–Q5 (units inside exponentials and cosines):** Quiz 4, Questions 1–2 (rt in e^{-rt}) and Quiz 5, Questions 1–2 (ωt and ω).
- **Q6 ($T - T_a$, C and the solution):** Quiz 3, Question 5 and Quiz 4, Questions 5–6 — the same equation with a hot object, so $C > 0$ there.
- **Q7 (the series argument):** Quiz 3, Question 6 and Quiz 4, Question 6 — the same factoring for e^{rt} and e^{-rt} , one derivative shallower.
- **Q8 (the period and the intern):** Quiz 5, Question 5 ($\omega^2 = g/L$, inverting a root) and Question 7 (units and precision of a reported result).
- **Q9–Q10:** Quiz 5, Question 7 (units of a fitted constant), and Question 6 above — the shift $T - T_a$ is the first-order version of $x - x_0$.

Still stuck? Bring it to office hours.